

YONAS ALEM

Senior Software Engineer | Backend & Full Stack Systems

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SUMMARY

Senior software engineer with 5+ years building and fixing production systems across payments, gaming, ecommerce, advertising, and government services. Cut infrastructure costs 35% (roughly \$100K per year), improved platform performance ~30%, and integrated payment rails handling more than 3 billion ETB in transaction volume. Works across the stack: PostgreSQL schema design and backend services through React frontends, AWS infrastructure, and production debugging.

PROFESSIONAL EXPERIENCE

Senior Full Stack Engineer — TPK Synergy Limited

Remote | December 2025 – Present

Amazon seller analytics platform covering marketplace analytics, FBA, finance, PPC, and inventory.

- Diagnosed a production failure where recurring sync jobs silently stopped running after each deploy; traced it through BullMQ job state, ioredis connections, Node.js process lifecycle, and Railway scheduler behavior, then added explicit shutdown handling across 17 cron processes.
- Integrated Amazon SP-API and Amazon Ads authentication end to end: OAuth authorization, LWA token exchange, credential management, onboarding state, and encrypted credential storage.
- Extended the platform to UK and EU marketplaces by threading marketplace and currency context through every service, including VAT-aware margin calculations and region-specific marketplace logic.
- Cut dashboard load times by moving expensive aggregations from application code into SQL and removing duplicate database queries.
- Replaced a drifting migration history with an idempotent, additive migration process after finding the production schema out of sync with the repository, making later schema changes safe to apply.
- Corrected financial and inventory figures by sourcing fees and costs from authoritative Amazon endpoints instead of incomplete downstream records.
- Built dashboard features including Top Sellers search and export, performance metrics, disbursement views, FBA shipment workflows, and multi-marketplace settings.

Senior Software Engineer — Mereb Technologies

Addis Ababa, Ethiopia | April 2022 – November 2025

Owned a production real-money gaming platform end to end: loyalty, payouts, fraud, and reporting.

- Improved platform performance ~30% through database indexing, read/write database separation, system-wide caching, pagination, transactional operations, and a redesign of the status-checking system.
- Reduced infrastructure costs ~35%, an estimated \$100K per year, through database, caching, and resource optimization.
- Cut report generation on ~1 million record datasets from 2 hours to under 30 minutes by reworking data access and report processing.
- Designed a gRPC and Protocol Buffers service architecture for new gaming features, including PHP and React clients, shared abstractions, and PHAR packaging for production.
- Built fraud monitoring that scored gameplay activity, balances, wins and losses, active days, and transaction behavior to flag suspicious deposit and withdrawal patterns.
- Automated game validation, reporting, and S3 cold-storage backups, cutting manual verification effort ~60%.
- Ran deployments and monitoring across test and production using Datadog, CloudWatch, EC2, load balancers, and proxy-based traffic management.
- Mentored 3 junior engineers on API development, debugging, code quality, and system design.

Full Stack Developer (Contract) — Affiliate.com

Remote | February 2023 – August 2023

- Migrated a legacy PHP API to Node.js and tRPC across campaigns, creatives, line items, P&L, affiliates, and dashboard functionality.

- Reimplemented complex legacy SQL behavior in Prisma while keeping business rules and data consistent between the legacy and new platforms running in parallel.
- Reduced report generation time ~40% by moving generation off the backend and into a Google Cloud Storage workflow.
- Rebuilt API validation in Zod without changing existing application behavior, and wrote unit and integration tests reaching ~85% coverage on migrated code.
- Built React and Tailwind CSS dashboards for reporting and data visualization.

Frontend Developer — Eaglelion Systems Technology

Addis Ababa, Ethiopia | February 2021 – April 2022

- Integrated Mastercard, Visa, Telebirr, and commercial bank payment APIs into products with 100K+ downloads and more than 3 billion ETB in transaction volume.
- Implemented payment workflows covering transaction states, callbacks, failures, retries, verification, and reconciliation across external providers.
- Replaced legacy SOAP integrations with REST APIs while keeping communication reliable against incomplete and inconsistent provider documentation.
- Raised the GTmetrix score from 45 to 97 and reduced page load times ~25% through code splitting, lazy loading, and asset optimization.
- Built responsive React and Next.js interfaces with Tailwind CSS and Material UI, connecting frontend applications to backend APIs and external services.

SELECTED PROJECTS

Government Constituent Platform — Autobridge Systems

- Designed database models and backend APIs for complaints, document approvals, and internal communication workflows.
- Traced multi-minute slowdowns on a table holding thousands of incoming email records to missing indexes; targeted indexing improved those operations by more than 10x.
- Implemented access controls and encryption for sensitive constituent data.
- Integrated AI services into communication workflows for threat detection and message classification.

Upplai — AI-Powered Resume Platform

- Integrated OpenAI and Gemini into resume generation, cover letter, and ATS feedback workflows supporting 500+ user profiles.
- Built resume scoring and document preview, plus per-resume cost tracking so model usage and spend were visible per generation.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, PHP, SQL

Backend: Node.js, Express, FastAPI, Prisma, gRPC, tRPC, REST, BullMQ

Databases: PostgreSQL, MySQL, MongoDB, Redis

Frontend: React, Next.js, Tailwind CSS, Material UI

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, Nginx, GitHub Actions, CI/CD, Datadog, CloudWatch

Integrations: Amazon SP-API, Amazon Ads API, Mastercard, Visa, Telebirr, OAuth 2.0, OpenAI, Gemini

Engineering: API Design, System Design, Microservices, Database Optimization, Performance Tuning, Testing, TDD, Code Review, Production Troubleshooting

EDUCATION & CERTIFICATION

BSc in Software Engineering — Addis Ababa University

Capstone: distributed system for real-time data processing.

AWS Certified Cloud Practitioner